Quick Reference Guide

Day 1 — Linux Administration Essentials

LNX-001

Day 1 · Modules 1, 2, 3, 4

How to use this guide: Essential on-the-job command reference for Ubuntu 24 system administration.

Task 1 — You need to search and filter system logs using pipelines and redirection

Locate entries: Filter log files for case-insensitive keywords using grep piped to pagers or summary counts. `grep -i` | `wc -l`

Direct stream output: Use single bracket redirection to overwrite target report files or double brackets to append updates. `>` `>>`

Follow live logs: Monitor incoming events in real time during troubleshooting sessions with the follow flag. `tail -f /var/log/syslog`

```
grep -i "error" /var/log/syslog | tail -n 20
grep "/bin/bash" /etc/passwd | wc -l > /tmp/bash_users_count.txt
```

Task 2 — You need to provision a new user account and assign supplementary group access

Create user identity: Provision the account with a home directory and interactive login shell in a single step. `useradd -m -s /bin/bash <user>`

Set credentials: Assign an initial password to the account to enable standard user authentication. `sudo passwd <user>`

Preserve memberships: Always use append mode when assigning secondary groups to prevent stripping existing group permissions. `usermod -aG <group> <user>`

Verify identity: Confirm assigned numerical identifiers and complete group memberships before releasing access. `id <user>`

```
sudo useradd -m -s /bin/bash arivera
sudo passwd arivera
sudo usermod -aG developers arivera
id arivera
```

Task 3 — You need to secure a shared directory with group ownership and boundary permissions

Assign directory ownership: Update both the owning user and owning group simultaneously using standard colon syntax. `chown <user>:<group> <path>`

Enforce group boundaries: Set octal mode 770 so group members have full access while unprivileged others are denied. `chmod 770 <path>`

Verify applied modes: Inspect target directory metadata to verify security tiers before placing collaborative files inside. `ls -ld <path>`

```
sudo chown -R arivera:developers /srv/projects
sudo chmod 770 /srv/projects
ls -ld /srv/projects
```

Octal Mode	POSIX String	Effective Access Rights
770	rwxrwx---	Full access for owner and group; no access for others
750	rwxr-x---	Full owner access, group read/execute, no access for others
640	rw-r-----	Owner read/write, group read-only, no access for others
644	rw-r--r--	Owner read/write, world read-only

Task 4 — You need to format a storage partition and configure persistent mounting in /etc/fstab

Format partition: Create an ext4 filesystem on the target block device partition using the standard filesystem builder. `mkfs.ext4 /dev/sdb1`

Retrieve filesystem UUID: Obtain the unique identifier to ensure persistent mounting independent of drive letter shifts. `sudo blkid /dev/sdb1`

Define fstab entry: Append the UUID definition specifying mount point, filesystem type, defaults options, dump, and fsck pass. `/etc/fstab`

Validate before reboot: Test mounting all configured filesystems immediately to catch syntax errors before the host reboots. `sudo mount -a`

```
sudo mkfs.ext4 /dev/sdb1
sudo blkid /dev/sdb1
# Example /etc/fstab entry: UUID=<UUID_VALUE> /mnt/data ext4 defaults 0 2
sudo mount -a
```

Task 5 — You need to locate and terminate an unresponsive background process safely

Locate process identifier: Find the active process ID using snapshot listing or pattern-matching tools. `ps aux | grep <name>` `pgrep -fl <name>`

Attempt graceful stop: Send termination signal 15 first so the application can flush buffers and close open locks cleanly. `kill -15 <PID>`

Force kill if stuck: Issue signal 9 only when the process ignores graceful termination and remains unresponsive. `kill -9 <PID>`

```
pgrep -fl worker_script
kill -15 <PID>
# If process fails to terminate after several seconds:
kill -9 <PID>
```

Task 6 — You need to configure host firewall rules with UFW without losing remote SSH access

Set default policies: Enforce baseline host defense by blocking all unsolicited incoming packets while allowing outbound connections. `ufw default deny incoming` `ufw default allow outgoing`

Allow SSH management: Explicitly permit OpenSSH connections before activating the firewall to prevent immediate administrative lockout. `sudo ufw allow OpenSSH`

Enable and verify: Turn on firewall packet filtering and review active rule tables in verbose mode. `sudo ufw enable` `sudo ufw status verbose`

```
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw allow OpenSSH
sudo ufw enable
sudo ufw status verbose
```